

Learning under Distribution Shift: Robust Classification of Gas Sensor Data under Temporal Drift

Davide Leone
davide.leone6@unibo.it

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1 Introduction

Machine learning models deployed in real-world systems frequently violate the standard assumption of stationary data distributions. Measurement systems evolve, sensors age, and operational environments shift, causing predictive models to suffer performance degradation over time. This challenge, commonly referred to as *distribution shift* or *concept drift*, is especially relevant in sensor-based applications such as chemical sensing arrays and High-Performance Computing (HPC) infrastructures. In HPC systems, which constitute the primary focus of my PhD research, monitoring signals including telemetry, power consumption, and temperature measurements evolve due to hardware aging, software updates, and changing workload characteristics. Understanding model behavior under temporal drift is therefore essential for building resilient predictive systems across engineering domains.

This study investigates model robustness using the Gas Sensor Array Drift Dataset, which contains observations collected from 16 chemical sensors across six gas classes over ten temporal batches. Although the dataset originates from environmental sensing, it exhibits the same fundamental challenge encountered in HPC monitoring systems: models are trained under one set of operating conditions and subsequently deployed on measurements generated by a continuously evolving system. As a result, the dataset provides a realistic testbed for studying the impact of temporal distribution shift on machine learning performance.

Rather than relying on standard random train-test splits, we leverage the chronological structure of the dataset to evaluate generalization under progressively increasing temporal distance from the training period. Five representative classification models are considered: Logistic Regression (LogReg), Support Vector Machines (SVM), Random Forests (RF), Extreme Gradient Boosting (XGB), and Multi-Layer Perceptrons (MLP). In addition to a static deployment scenario, where models are trained once and evaluated on future batches,

we investigate an adaptive learning strategy based on periodic retraining. This allows us to assess whether updating the training distribution is more effective than increasing model complexity when dealing with non-stationary environments.

The study addresses three main research questions:

- **RQ1:** How does temporal distribution shift impact classification performance, and which machine learning models are most robust when trained only on historical data?
- **RQ2:** Is performance degradation primarily driven by the overall magnitude of distribution shift or by localized changes in class separability?
- **RQ3:** Can periodic retraining effectively mitigate sensor drift and maintain long-term predictive robustness?

The experimental results show that temporal degradation cannot be explained solely by the global displacement of data distributions. Instead, model performance depends strongly on how drift affects local class structure and decision boundaries. Furthermore, the dataset exhibits not only feature drift caused by evolving sensor responses, but also substantial changes in class frequencies across batches, introducing an additional form of distribution shift known as *prior probability shift*. Together, these effects produce complex degradation patterns that cannot be captured by a single drift metric. Finally, the results demonstrate that periodic retraining consistently improves robustness across all evaluated architectures, suggesting that maintaining an up-to-date training distribution is often more important than increasing model complexity when operating in evolving environments.

2 Dataset and Experimental Setup

2.1 Dataset and Temporal Evaluation Protocol

This study utilizes the Gas Sensor Array Drift Dataset, a standard benchmark designed for evaluating classification models under temporal distribution shift. The dataset comprises 13,910 measurements gathered from 16 chemical sensors exposed to six distinct gases across ten chronologically ordered batches. Table 1 outlines the overall dataset composition, showing that the aggregate dataset is relatively balanced.

To rigorously assess performance under distribution shift, we adopt a chronological splitting protocol instead of a random train-test split. Batches 1–5 serve as the historical training set, while Batches 6–10 are used as sequential test sets. Evaluating each test batch individually allows direct tracking of performance degradation as temporal distance from the training window grows. In addition to feature space variations over time, the chronological split exposes significant fluctuations in class frequencies across individual acquisition batches. As shown in Table 2, class proportions fluctuate significantly: Gas 6 constitutes only 2.2%

Gas Type	Gas Name	Samples
1	Ethanol	2,565
2	Ethylene	2,926
3	Ammonia	1,641
4	Acetaldehyde	1,936
5	Acetone	3,009
6	Toluene	1,833

Table 1: Overall distribution of gas types in the dataset.

of the initial training set (Batches 1–5) but jumps to over 20% in Batches 6 and 9; conversely, Gas 4 represents 12.5% of the training set but drops to just 1.3% in Batch 6. These temporal variations suggest that predictive degradation in real-world deployments cannot be attributed solely to feature drift, but may also be driven by underlying prior probability shift.

Dataset	Gas 1	Gas 2	Gas 3	Gas 4	Gas 5	Gas 6
Train (1–5)	0.196	0.277	0.119	0.125	0.262	0.022
Batch 6	0.223	0.250	0.048	0.013	0.263	0.203
Batch 7	0.180	0.183	0.100	0.206	0.174	0.157
Batch 8	0.102	0.102	0.136	0.112	0.486	0.061
Batch 9	0.130	0.117	0.213	0.160	0.166	0.215
Batch 10	0.167	0.167	0.167	0.167	0.167	0.167

Table 2: Normalized class distributions across training and testing batches.

2.2 Models and Learning Paradigms

To systematically investigate how different machine learning paradigms handle temporal distribution shift, we evaluate five representative models ranging from linear baselines to non-linear kernel methods, tree ensembles, and neural networks:

- **Logistic Regression (LogReg):** A linear discriminative classifier that estimates class probabilities through a softmax transformation. It serves as an interpretable baseline and allows us to evaluate how well simple linear decision boundaries generalize under temporal drift.
- **Support Vector Machine (SVM):** A margin-based classifier using a Radial Basis Function (RBF) kernel. By implicitly projecting samples into a higher-dimensional feature space, SVMs can model complex non-linear relationships while maintaining strong regularization properties.
- **Random Forest (RF):** An ensemble learning method that combines multiple decision trees trained on bootstrap samples. Random Forests

can capture highly non-linear decision boundaries and interactions among sensor features while reducing variance through averaging.

- **Extreme Gradient Boosting (XGB):** A gradient boosting framework that sequentially constructs decision trees to correct the errors of previous trees. XGBoost is widely regarded as one of the strongest methods for tabular data and serves as a representative boosting-based approach.
- **Multi-Layer Perceptron (MLP):** A feedforward neural network with two hidden layers using ReLU activation functions. The MLP provides a flexible non-linear model capable of learning complex feature representations directly from sensor readings.

The objective of this study is not to identify the best-performing configuration for each classifier, but rather to compare their relative robustness under distribution shift. Consequently, a single set of reasonable hyperparameters (reported in Table 3) was selected for each model and kept fixed throughout both the static and adaptive experiments. This design choice ensures that observed performance differences are attributable primarily to the learning paradigm and the effects of temporal drift rather than repeated hyperparameter optimization. Moreover, the dataset contains only ten temporal batches, and allocating additional batches for tuning would reduce the amount of data available for drift analysis.

Model	Main Hyperparameters
Logistic Regression	max_iter = 5000
SVM	RBF kernel, $C = 10$
Random Forest	300 trees
XGBoost	300 trees, max_depth = 6
MLP	Hidden layers = (128, 64), ReLU

Table 3: Model hyperparameter configurations.

Prior to model training, all features are standardized using statistics computed exclusively from the training data. The same transformation is subsequently applied to all future batches to avoid information leakage.

To assess how operational strategies mitigate performance loss over time, we compare two distinct operational paradigms:

- **Static Learning:** Reflects a standard baseline deployment scenario. Models are trained once on historical data (Batches 1–5) and subsequently evaluated across test sets (Batches 6–10) without any parameter updates or retraining.
- **Adaptive Learning:** Simulates a realistic continuous learning scenario where models are periodically retrained as new historical batches are collected. Formally, when predicting target batch $k \in \{6, \dots, 10\}$, the model

is trained on all accumulated prior data $\bigcup_{i=1}^{k-1} \text{Batch}_i$. For example, Batch 7 is predicted using a model retrained on Batches 1–6, and Batch 10 is predicted using Batches 1–9. Importantly, when predicting Batch k , only data from batches preceding k are available. The target batch itself is never included in the training set, thereby preserving the chronological integrity of the evaluation.

2.3 Evaluation Metrics

Classification performance is evaluated using three standard metrics: Accuracy (total number of correct predictions divided by the total number of samples), Balanced Accuracy (average accuracy across classes), and Macro F1 (average of the F1-scores across classes). Balanced Accuracy and Macro F1 are particularly essential here to prevent dominant classes from hiding poor performance on under-represented gas types.

To summarize model behavior across the entire evaluation trajectory, we define two aggregate temporal metrics:

- **Performance Degradation (Deg):** Quantifies the net change between the initial and final test batches:

$$\text{Deg}(M) = \text{Metric}_{\text{Batch } 10} - \text{Metric}_{\text{Batch } 6}$$

where negative values denote performance decay over time. This metric captures the cumulative performance loss between the first and last evaluation periods and therefore reflects long-term degradation rather than short-term fluctuations.

- **Robustness (Rob):** Measures the mean performance maintained across all $N = 5$ testing periods:

$$\text{Rob}(M) = \frac{1}{N} \sum_{b=6}^{10} \text{Metric}_b$$

This metric captures the performance maintained over time, with high robustness indicating that a classifier consistently retains strong predictive power across shifting regimes, even if short-term drops occur.

3 Distribution Shift Analysis

This section analyzes the behavior of machine learning models under the static learning paradigm, where training is restricted to the historical batches (1–5) and no model updates are performed after deployment. First, we evaluate how predictive performance evolves as the temporal distance from the training period increases (RQ1). Then, we investigate whether performance degradation is directly explained by the magnitude of distribution shift or whether other factors, such as class separability and changing class proportions, play a more important role (RQ2).

3.1 Performance under Static Learning

Table 4 details the sequential classification performance across all five architectures .

To establish a baseline for temporal degradation, Table 4 details the classification performance of all five models over the five test batches, while Figure 1 illustrates their performance trajectories across Accuracy, Balanced Accuracy, and Macro F1

Model	Metric	B6	B7	B8	B9	B10
LogReg	Accuracy	0.776	0.648	0.510	0.617	0.475
	Bal. Acc.	0.808	0.661	0.535	0.659	0.475
	Macro F1	0.658	0.596	0.501	0.574	0.424
SVM	Accuracy	0.740	0.741	0.806	0.760	0.456
	Bal. Acc.	0.783	0.737	0.743	0.807	0.456
	Macro F1	0.633	0.693	0.686	0.737	0.444
RF	Accuracy	0.713	0.579	0.190	0.562	0.364
	Bal. Acc.	0.766	0.592	0.220	0.574	0.364
	Macro F1	0.625	0.535	0.107	0.536	0.352
XGB	Accuracy	0.620	0.577	0.320	0.557	0.390
	Bal. Acc.	0.696	0.588	0.356	0.553	0.390
	Macro F1	0.575	0.550	0.274	0.454	0.370
MLP	Accuracy	0.760	0.661	0.776	0.530	0.501
	Bal. Acc.	0.798	0.674	0.612	0.525	0.501
	Macro F1	0.697	0.612	0.573	0.443	0.452

Table 4: Classification performance under static learning across Batches 6–10.

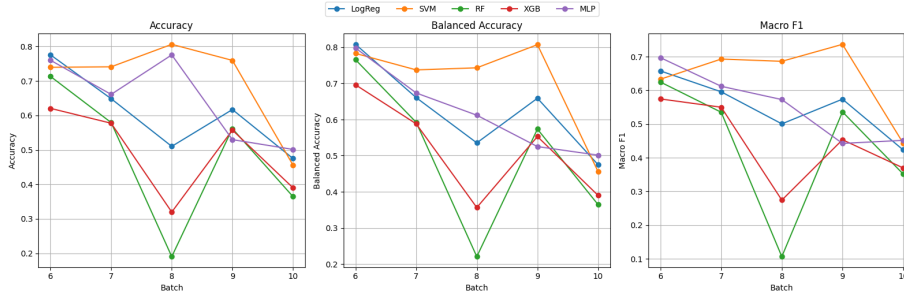


Figure 1: Evolution of Accuracy, Balanced Accuracy, and Macro F1 under static deployment.

The empirical evaluation reveals three key trends:

- All evaluated models exhibit severe long-term decay between the initial evaluation point (Batch 6) and the final period (Batch 10), confirming that sensor drift fundamentally degrades static decision boundaries over extended horizons.

- Performance degradation is non-monotonic across evaluation windows. Both SVM and MLP exhibit substantial temporary recoveries on Batches 8 and 9 relative to Batch 7. For example, SVM Accuracy peaks at 0.806 in Batch 8. This observation suggests that the difficulty of a target batch depends not only on how far it drifts from the training distribution but also on how the drift affects class overlap and class proportions.
- Tree ensembles experience severe performance drops on Batch 8. Random Forest accuracy drops to 0.190 (Macro F1 = 0.107), while XGBoost drops to 0.320 (Macro F1 = 0.274). One possible explanation is that tree-based models partition the feature space via orthogonal, axis-aligned hyperplanes, so they lack extrapolation capabilities outside historical training bounds. When sensor readings shift into previously unseen numeric ranges, tree leaves blindly assign samples to edge boundaries.

To rigorously summarize long-term stability and overall operational robustness, Tables 5 and 6 present the cumulative degradation (Deg) and mean robustness (Rob) metrics.

Model	Accuracy Deg	Bal. Acc. Deg	Macro F1 Deg
LogReg	-0.301	-0.333	-0.234
SVM	-0.284	-0.327	-0.190
RF	-0.349	-0.401	-0.273
XGB	-0.230	-0.306	-0.205
MLP	-0.259	-0.297	-0.245

Table 5: Performance degradation between Batch 6 and Batch 10.

Model	Accuracy Rob	Bal. Acc. Rob	Macro F1 Rob
LogReg	0.605	0.628	0.550
SVM	0.700	0.705	0.639
RF	0.482	0.503	0.431
XGB	0.493	0.517	0.444
MLP	0.646	0.622	0.555

Table 6: Mean robustness scores maintained across Batches 6–10.

Quantitatively, Random Forest exhibits the highest overall degradation across all metrics ($\text{Deg}_{\text{BalAcc}} = -0.401$, $\text{Deg}_{\text{F1}} = -0.273$). Conversely, SVM delivers the highest robustness across the deployment window ($\text{Rob}_{\text{Acc}} = 0.700$, $\text{Rob}_{\text{F1}} = 0.639$), outperforming tree ensembles by over 20 percentage points in Macro F1. A possible explanation is that the smooth Radial Basis Function (RBF) kernel boundaries of the SVM provide superior generalization under static feature shift.

3.2 Quantifying Global Temporal Drift

To investigate whether performance degradation correlates directly with feature space displacement, Principal Component Analysis (PCA) was fitted on the training baseline (Batches 1–5). Target batch centroids were then projected into this latent space. Figure 2 displays the centroid trajectories, along with the Euclidean distances relative to the training baseline.

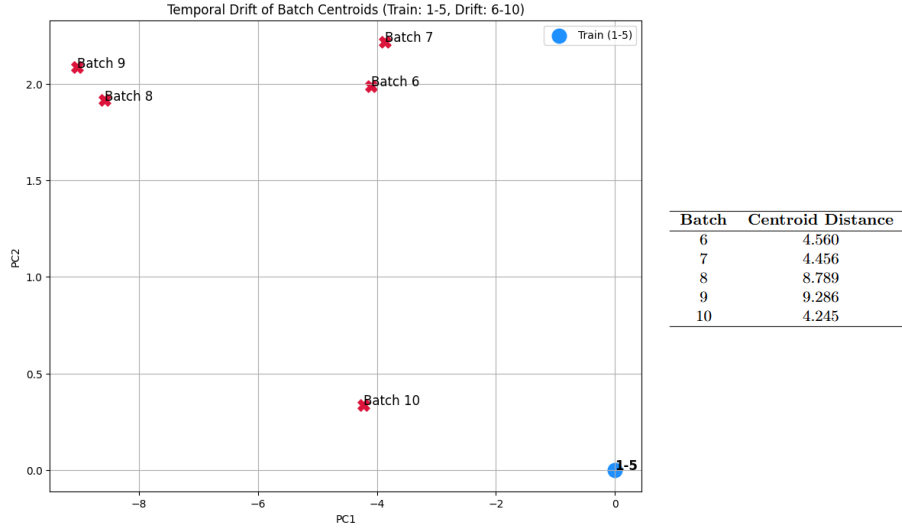


Figure 2: Centroid trajectory and Euclidean distances of test batches in PCA space relative to the training baseline (Batches 1–5).

A clear contradiction emerges when comparing global centroid displacement with model classification performance. Batches 8 and 9 exhibit the largest centroid distances (8.789 and 9.286), yet SVM retain high classification accuracy (0.806 and 0.760). Conversely, Batch 10 records the smallest centroid distance (4.245), despite producing the lowest accuracy across all models. This result indicates that global distribution shift metrics alone are insufficient predictors of model degradation. If classification performance were governed primarily by overall displacement in feature space, Batch 10 would be expected to be easier than Batches 8 and 9. The opposite behavior is observed, suggesting that the internal structure of the drift is more important than its aggregate magnitude.

3.3 Visual Analysis of Latent Feature Drift

To resolve this paradox, Figure 3 projects the individual sample distributions of each test batch against the baseline support (grey markers).

The visual projections reveal the underlying feature distribution dynamics:

- **Batches 6 and 7:** Sample distributions remain largely aligned with

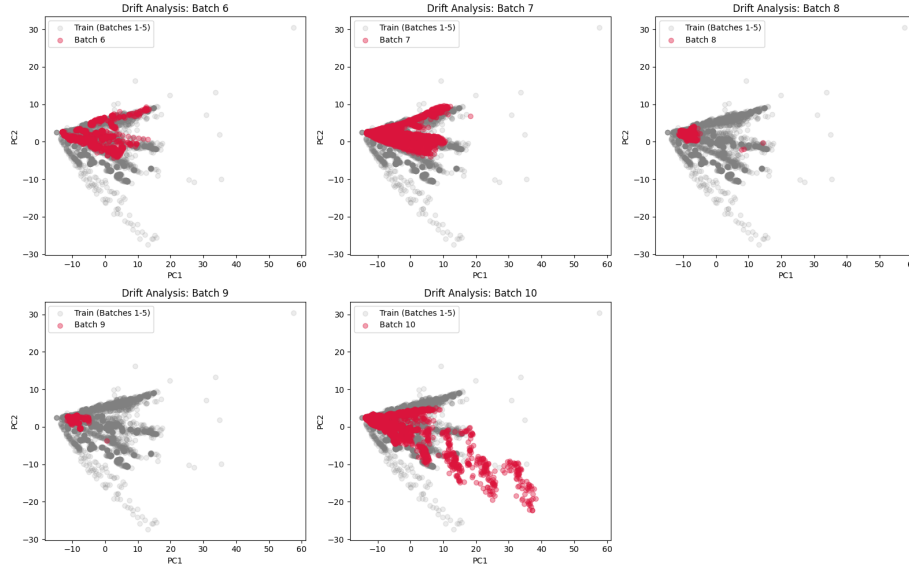


Figure 3: Projection of test batches (red) onto the PCA space defined by the training baseline (grey).

the central density regions of the baseline, preserving historical decision boundaries.

- **Batches 8 and 9:** Samples do not exit the support boundary of the training data. Instead, observations become tightly concentrated within a localized sub-region along the left boundary. This localized spatial concentration pulls the batch centroid significantly to the left, generating large aggregate Euclidean distances. However, because these points reside inside historical support boundaries, non-linear classifiers with smooth boundaries (like SVM) successfully discriminate classes.
- **Batch 10:** Although the aggregate centroid remains close to the historical center (4.245), individual observations expand downward along PC2. This shift forces samples into previously unobserved regions of feature space and disrupts historical decision boundaries, resulting in severe accuracy loss.

3.4 Class-Level Drift and Class Distribution Shift

To analyze class-conditional dynamics, Figure 4 resolves the latent feature projections into individual gas classes across each evaluation period.

Class-level projections confirm that temporal drift affects gas classes non-uniformly:

- **Batches 6 and 7:** Samples remain broadly distributed across the historical feature space, closely tracking the baseline training manifold geometry

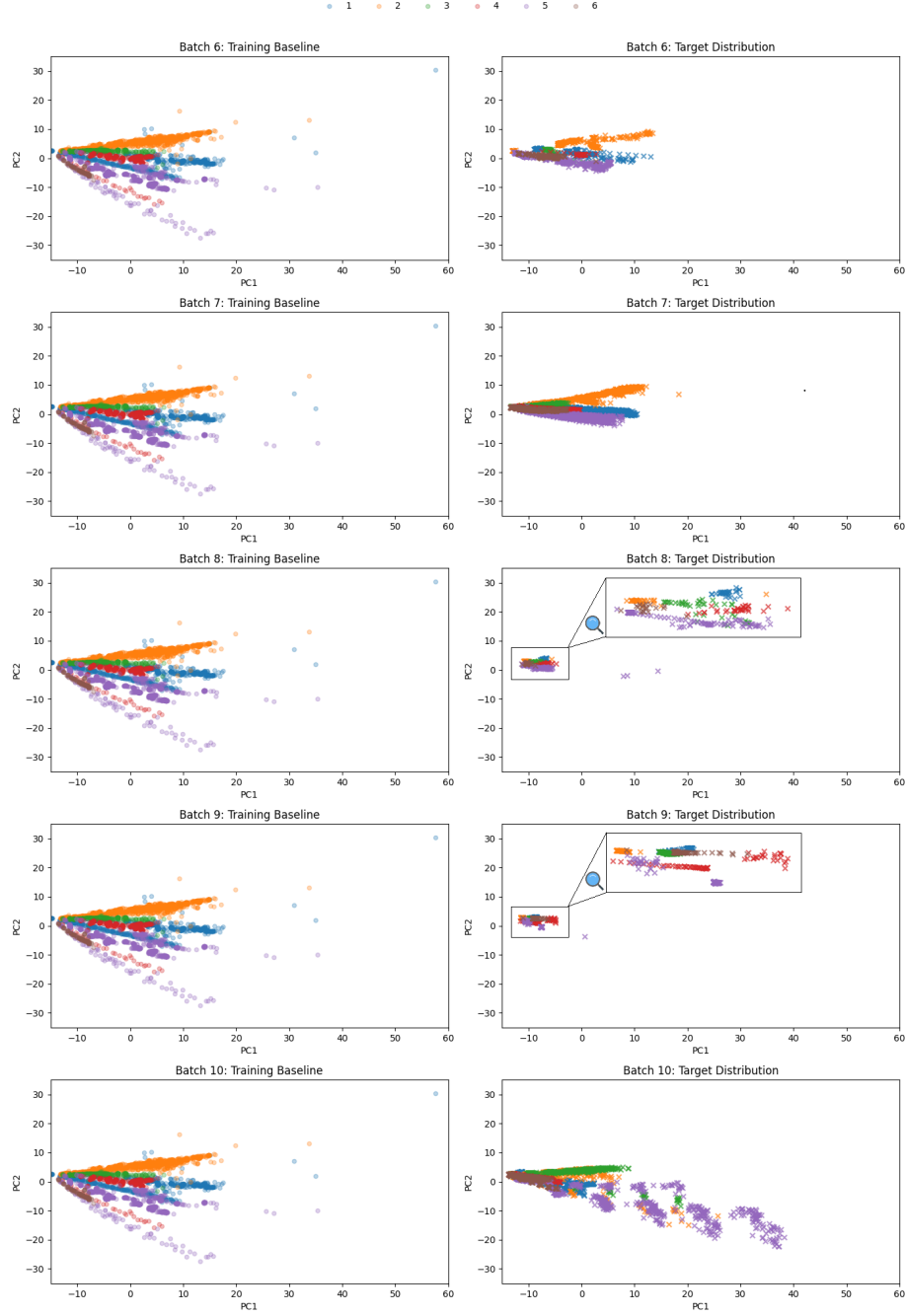


Figure 4: Class-specific drift analysis in PCA space. Left panels display the baseline training distribution per batch row; right panels display target batch samples colored by gas class.

along both PC1 and PC2. Feature drift is minimal during this early period, preserving fundamental spatial boundaries between classes.

- **Batches 8 and 9:** Target samples undergo severe spatial compression, becoming strictly localized within a narrow window. Crucially, comparing the target distributions to the baseline training manifold within this window demonstrates that target observations occupy similar local sub-manifolds established during training. The classes either remain strictly ordered along distinct parallel bands or form a clear, contiguous segment. Because target samples remain strictly within historical training support and preserve clear class-conditional margins, non-linear classifiers with continuous, smooth decision boundaries (SVM with RBF kernel) maintain exceptional accuracy. Conversely, decision tree ensembles (RF, XGB) probably collapse because their axis-aligned orthogonal cuts cannot adapt to high spatial sample concentration within such narrow local feature bounds
- **Batch 10:** Structural drift manifests as two distinct distributional transformations. First, Gas 5 (purple) undergoes unprecedented spatial expansion, moving far to the right along PC1 and downward along PC2, disrupts historical decision boundaries of Gas 5. Second, near the historical origin, Gas 6 (brown) and Gas 4 (red) collapse directly onto the support region occupied by Gas 1 (blue). Notably, this severe manifold overlap in Batch 10 corresponds directly to the misclassification patterns observed in the confusion matrix analysis (Section 3.5), where Gas 6 and Gas 4 suffer massive misclassification into Gas 1. This alignment between geometric class overlap and empirical error patterns confirms that performance collapse is driven by a fundamental loss of local class separability rather than simple global centroid displacement.

In addition to feature drift, static performance is impacted by *prior probability shift*. As established in Table 2, class proportions fluctuate significantly over time: Gas 6 accounts for only 2.2% of the training data but expands to over 20% in Batch 6 and Batch 9. In contrast, Batch 8 is heavily biased toward Gas 5 (48.6%). Consequently, unweighted overall Accuracy can mask class-specific performance drops, reinforcing the necessity of evaluating Balanced Accuracy and Macro F1.

3.5 Static Error Diagnosis

To examine specific failure mechanisms, Figure 5 compares normalized confusion matrices for the top-performing static model (SVM) on Batch 6 and Batch 10.

The error patterns illustrate the transition from localized prior shift to severe boundary collapse:

- **Batch 6 (Figure 5a):** Classes 1–5 exhibit high diagonal recall (0.91–1.00). However, Class 6 yields 0.00 recall, with 0.92 of its samples misclassified as

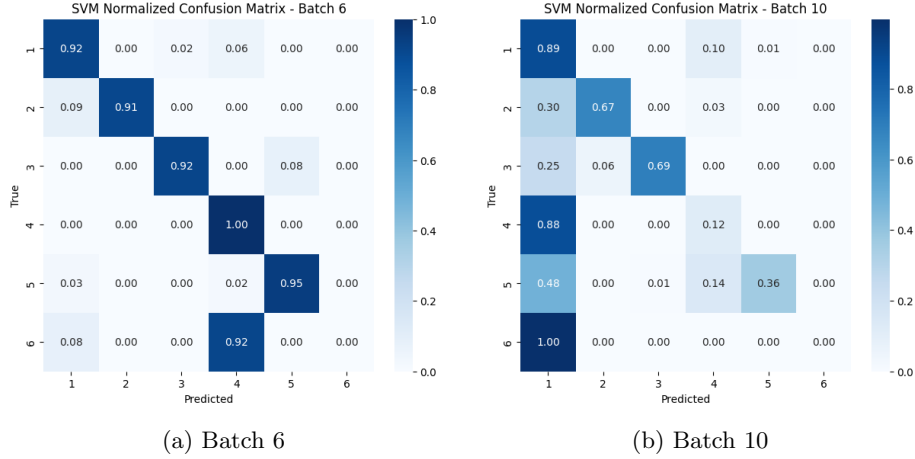


Figure 5: Normalized confusion matrices of the static SVM classifier on Batch 6 and Batch 10.

Class 4. This failure illustrates the impact of prior probability shift. Because the static model has observed relatively few examples of this class during training (2.2%), it struggles to recognize it when its prevalence increases substantially (20.3%).

- **Batch 10 (Figure 5b):** Class discrimination collapses across multiple classes. Class 6 experiences 100% misclassification into Class 1. Similarly, 88% of Class 4 samples are misclassified as Class 1.

Overall, the combined evidence from performance metrics, PCA projections, class-level visualizations, and confusion matrices reveals that temporal drift affects the dataset through multiple mechanisms simultaneously. Feature distributions move over time, class proportions change across batches, and the relative positioning of classes evolves. These factors interact to determine whether a classifier remains robust or experiences severe degradation.

Answer to RQ1. Temporal distribution shift causes substantial degradation across all static classifiers. SVM demonstrates the highest overall robustness ($\text{Rob}_{F1} = 0.639$), whereas tree ensembles (Random Forest, XGBoost) suffer severe instability, probably due to rigid axis-aligned decision boundaries.

Answer to RQ2. Performance degradation is governed by local class separability and prior probability shifts rather than the magnitude of global centroid displacement. Large centroid displacement within historical training support (Batches 8–9) preserves classification boundaries, whereas unobserved feature expansion into overlapping class regions (Batch 10) causes severe decision boundary collapse.

4 Adaptive Learning under Temporal Drift

While static learning established that temporal drift severely degrades classification performance over time, this section addresses the efficacy of periodic adaptive retraining in mitigating temporal drift and restoring classification accuracy (**RQ3**).

In the adaptive setting, classifiers are retrained prior to evaluating each target batch k using all previously observed batches (1 through $k - 1$). Model architectures and hyperparameters remain identical to the static experiments to isolate the effect of continuous data adaptation.

4.1 Performance under Adaptive Retraining

Table 7 details the classification metrics across the sequential test batches under adaptive retraining, while Figure 6 tracks these trajectories over time.

Model	Metric	B6	B7	B8	B9	B10
LogReg	Accuracy	0.776	0.833	0.918	0.770	0.715
	Bal. Acc.	0.808	0.850	0.894	0.817	0.715
	Macro F1	0.658	0.826	0.890	0.756	0.707
SVM	Accuracy	0.740	0.877	0.915	0.772	0.730
	Bal. Acc.	0.783	0.887	0.878	0.820	0.730
	Macro F1	0.633	0.878	0.900	0.751	0.728
RF	Accuracy	0.713	0.754	0.871	0.843	0.610
	Bal. Acc.	0.766	0.763	0.747	0.876	0.610
	Macro F1	0.625	0.721	0.742	0.852	0.609
XGB	Accuracy	0.620	0.728	0.830	0.751	0.607
	Bal. Acc.	0.696	0.736	0.742	0.786	0.607
	Macro F1	0.575	0.688	0.712	0.743	0.610
MLP	Accuracy	0.760	0.791	0.908	0.768	0.720
	Bal. Acc.	0.798	0.810	0.874	0.816	0.720
	Macro F1	0.697	0.775	0.892	0.745	0.727

Table 7: Classification performance under adaptive learning across Batches 6–10.

Adaptive retraining yields dramatic stability and performance gains across all classifiers:

- Incorporating previous batches into the training set improves performance substantially. Already with Batch 7, SVM Macro F1 jumps from 0.693 (static) to 0.878 (adaptive). Peak performance occurs at Batch 8, where Logistic Regression, SVM, and MLP all exceed 0.90 Accuracy and 0.88 Macro F1.
- Random Forest and XGBoost, which suffered catastrophic failure on Batch 8 under static learning (RF F1: 0.107), achieve Macro F1 scores of 0.742 and 0.712 under adaptive retraining, reaching their peak performance on

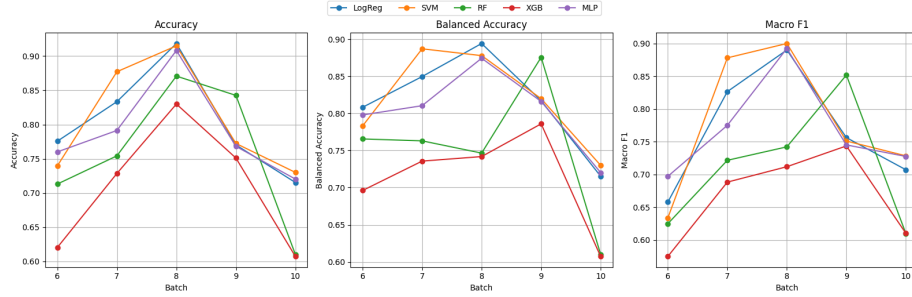


Figure 6: Evolution of Accuracy, Balanced Accuracy, and Macro F1 under adaptive learning.

Batch 9 (RF F1: 0.852). By retraining tree splits on expanded numerical bounds, adaptive learning resolves boundary mismatches outside historical training regions.

- The performance gap between complex non-linear models and linear classifiers narrows significantly. Logistic Regression achieves robust predictive accuracy ($\text{Rob}_{F1} = 0.767$), performing competitively with SVM (0.778) and MLP (0.767). Re-estimating linear hyperplanes on recent data effectively tracks moving class centroids.

Tables 8 and 9 present the adaptive degradation (Deg) and robustness (Rob) metrics.

Model	Accuracy Deg	Bal. Acc. Deg	Macro F1 Deg
LogReg	-0.060	-0.093	+0.049
SVM	-0.010	-0.053	+0.095
RF	-0.103	-0.156	-0.015
XGB	-0.013	-0.089	+0.035
MLP	-0.040	-0.078	+0.030

Table 8: Performance degradation between Batch 6 and Batch 10 under adaptive learning.

Measured relative to Batch 6 initial values, terminal degradation is effectively mitigated, with SVM, LogReg, XGB, and MLP exhibiting positive Macro F1 degradation ($\text{Deg}_{F1} > 0$). Overall robustness increases across all models (Rob_{F1} rises from 0.639 to 0.778 for SVM, and from 0.431 to 0.710 for Random Forest).

However, **adaptive retraining does not eliminate absolute performance decay relative to peak deployment windows**. The degradation metric defined in Section 2.3 measures the difference between Batch 6 and Batch 10. Under adaptive learning this quantity becomes very small, and in some cases positive for Macro F1. However, this does not imply that performance remains constant throughout deployment.

Model	Accuracy Rob	Bal. Acc. Rob	Macro F1 Rob
SVM	0.807	0.820	0.778
LogReg	0.803	0.817	0.767
MLP	0.790	0.804	0.767
RF	0.758	0.752	0.710
XGB	0.707	0.713	0.666

Table 9: Mean robustness scores maintained across Batches 6–10 under adaptive learning.

In fact, most models achieve their highest performance on Batches 8 or 9 before experiencing a noticeable decline on Batch 10. For example, SVM Accuracy decreases from 0.915 on Batch 8 to 0.730 on Batch 10. Therefore, adaptive retraining substantially reduces long-term degradation relative to the initial deployment period, but it does not completely eliminate the effects of severe future drift.

4.2 Geometric Trajectory of Adaptive Baselines

Figure 7 illustrates the trajectory of the moving training centroid (Train 1– k) relative to individual target batch centroids in PCA space.

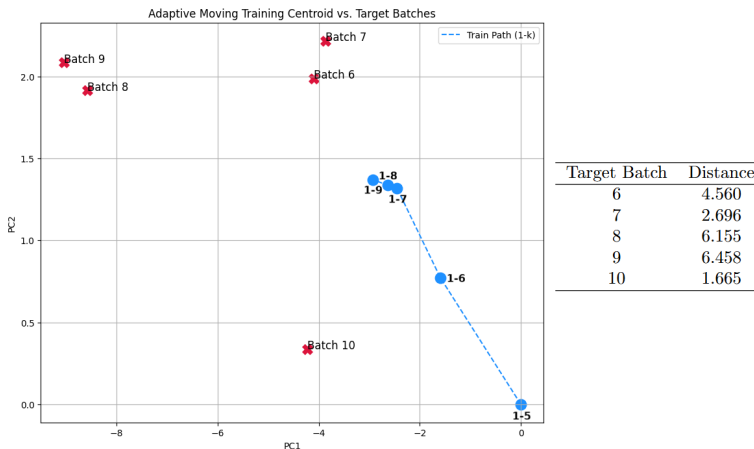


Figure 7: Adaptive moving training centroid trajectory vs. target batch centroids in PCA space, with corresponding Euclidean distances.

As additional batches are incorporated, the cumulative training centroid migrates smoothly from (0,0) (Batches 1–5) toward (−3.0,1.4) (Batches 1–9). Consequently, the Euclidean distance between the training baseline and target batches drops significantly compared to the static setting. For example, Batch 8 centroid distance decreases from 8.789 (static) to 6.155 (adaptive), while Batch

10 centroid distance decreases from 4.245 (static) to 1.665 (adaptive). This shift confirms that continuous data accumulation aligns the geometric center of the training manifold with the evolving sensor response space.

4.3 Visual Analysis of Expanding Training Support

Figure 8 projects target batch observations (red) against the cumulative training support (grey) as the training set expands from Batches 1–5 to Batches 1–9.

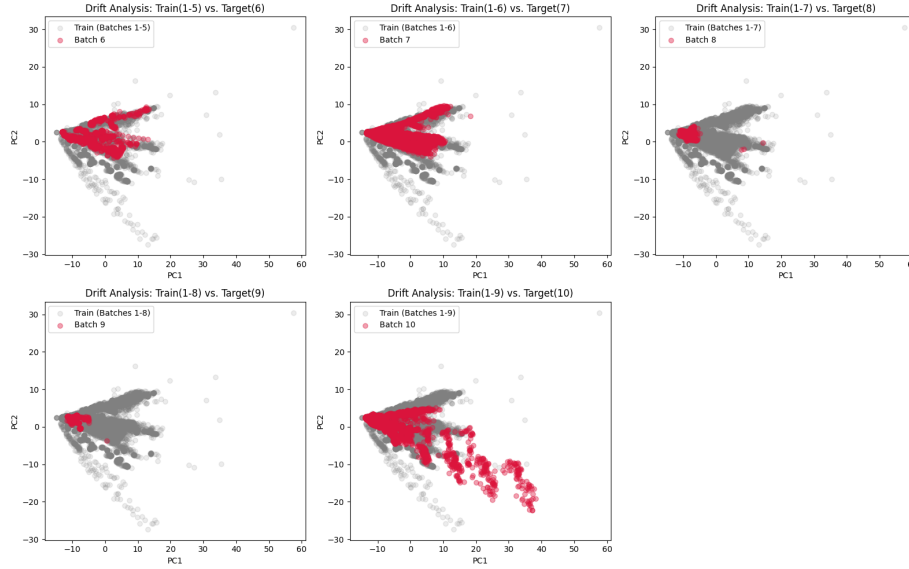


Figure 8: PCA projection of target batches (red) against the progressively expanding cumulative training baseline (grey).

The visual projections illustrate how adaptive learning progressively aligns the training distribution with the target distribution:

- **Batches 8 and 9:** Retraining on Batches 1–7 expands the cumulative baseline (grey markers) into the region occupied by observations in Batches 8 and 9, providing more density coverage for their classification.
- **Batch 10:** Incorporating Batches 6–9 expands historical training support downward, covering a much broader region. However, a significant portion of Batch 10 samples still extends beyond the lower boundary of Batches 1–9, explaining why Batch 10 performance degrades relative to Batch 8 even under adaptive learning.

Overall, the PCA projections suggest that future batches become increasingly represented within the support of the training data, which is consistent with the observed improvements in classification performance.

4.4 Prior Probability Alignment

Adaptive retraining simultaneously mitigates feature drift and prior probability shift. Table 10 compares the class distributions across training configurations.

Dataset	Gas 1	Gas 2	Gas 3	Gas 4	Gas 5	Gas 6
Train (1–5)	0.196	0.277	0.119	0.125	0.262	0.022
Train (1–9)	0.191	0.226	0.101	0.130	0.234	0.120
Batch 10	0.167	0.167	0.167	0.167	0.167	0.167

Table 10: Class distribution shift between static training (1–5), adaptive training (1–9), and Batch 10.

In the static baseline (1–5), Gas 6 represented only 2.2% of observations, causing models to collapse decision boundaries for Gas 6. In the adaptive training set (1–9), Gas 6 representation expands to 12.0%, aligning more closely with the uniform distribution of Batch 10 (16.7% per class). This change reduces the mismatch between training and deployment priors. In the static setting, models were trained under the assumption that Gas 6 was extremely rare. When Gas 6 became substantially more frequent in later batches, decision boundaries were poorly calibrated for its occurrence. By incorporating Batches 6–9 into the training process, adaptive learning partially restores class balance and exposes the models to a broader range of Gas 6 observations.

4.5 Adaptive Error Diagnosis

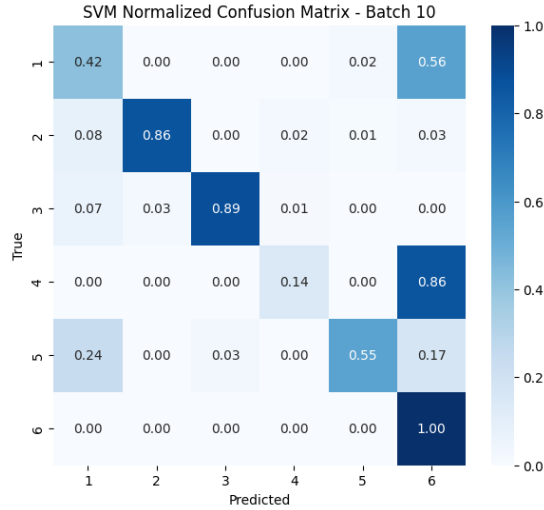


Figure 9: Normalized confusion matrix of the adaptively retrained SVM on Batch 10.

Figure 9 displays the normalized confusion matrix for the adaptively re-trained SVM on Batch 10. Comparing Figure 9 with the static confusion matrix (Figure 5b) reveals fundamental changes in model behavior:

- Adaptive learning significantly improves classification accuracy of Gas 2 (from 0.67 to 0.86) and Gas 3 (from 0.69 to 0.89).
- Under static learning, Gas 6 suffered 100% misclassification into Gas 1. Under adaptive learning, Gas 6 achieves 100% correct classification. Incorporating Batches 6–9 provides sufficient prior weight and spatial support to construct accurate boundaries for Gas 6. This result demonstrates that part of the performance degradation observed under static learning was not solely caused by feature drift, but also by the severe mismatch between historical and future class distributions.
- Under static learning, Gas 1, 4, and 5 were all frequently classified as Gas 1, while under adaptive learning, they are still frequently misclassified, this time as Gas 6. To understand this error structure we can look at latent feature drift in Figure 10. One possible explanation is that the increased representation of Gas 6 in the adaptive training set alters the learned decision regions in areas where classes overlap. Consequently, ambiguous samples that were previously assigned to other classes may now be classified as Gas 6. Thus, while adaptive retraining eliminates class collapse, severe physical overlap in unobserved feature space limits total accuracy.

4.6 Static vs Adaptive Learning

Model	Static Deployment	Adaptive Deployment
	Macro F1 Rob	Macro F1 Rob
SVM	0.639	0.778
LogReg	0.550	0.767
MLP	0.555	0.767
RF	0.431	0.710
XGB	0.444	0.666

Table 11: Comparison of mean robustness scores maintained across Batches 6–10 under static and adaptive deployment.

Table 11 compares the robustness scores obtained under static and adaptive deployment.

Adaptive retraining improves robustness across every evaluated architecture. The largest gains are observed for Random Forest, whose Macro F1 robustness increases from 0.431 to 0.710 (+64.7%), and XGBoost, which improves from

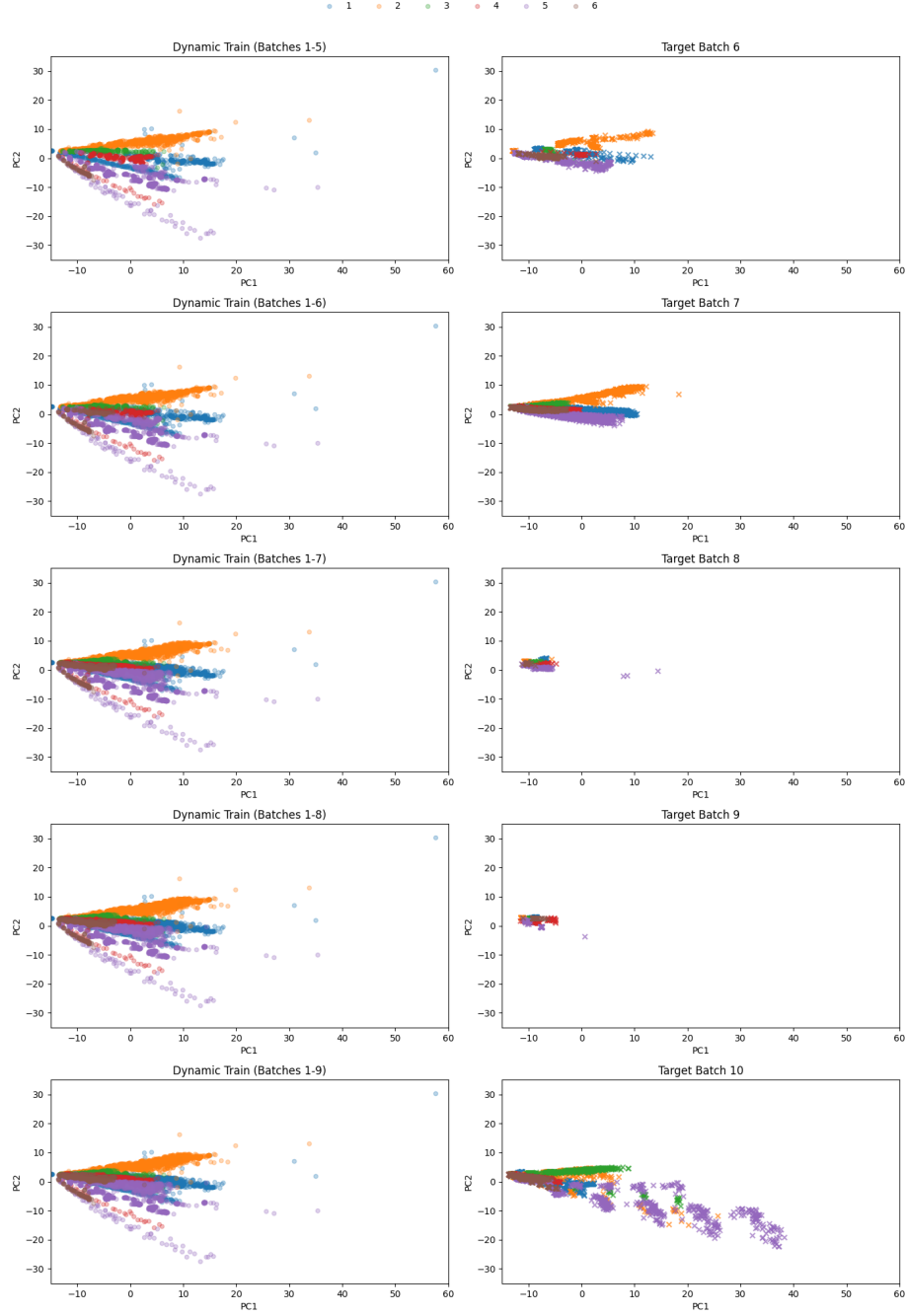


Figure 10: Class-specific PCA distribution under adaptive learning. Left column shows expanding cumulative training sets; right column shows corresponding target batch class distributions.

0.444 to 0.666 (+49.8%). Even the strongest static model, SVM, benefits substantially, with Macro F1 robustness increasing from 0.639 to 0.778 (+21.8%).

These results indicate that maintaining an up-to-date training distribution is more important than increasing model complexity alone. Under static deployment, the simple Logistic Regression model eventually outperforms both tree-based ensembles on several batches. After adaptation, performance differences between architectures become considerably smaller, suggesting that access to recent data contributes more to long-term robustness than the specific classifier family.

Answer to RQ3. Periodic adaptive retraining effectively mitigates temporal sensor drift, increasing overall Macro F1 robustness from 0.639 to 0.778 for SVM, and from 0.431 to 0.710 for Random Forest. Retraining enables even simple linear models such as Logistic Regression to achieve robustness comparable to considerably more sophisticated architectures. This suggests that access to recent representative data can be more important than model complexity when operating under temporal distribution shift. Performance recovery relies on dual mechanisms: **feature space alignment** (expanding training support into previously unobserved spatial clusters) and **prior probability adjustment** (rebalancing class representation, such as expanding Gas 6 from 2.2% to 12.0%). However, adaptation cannot prevent degradation when target samples expand into unobserved regions where distinct gas classes physically overlap in feature space.

5 Conclusions

This work investigated the impact of temporal distribution shift on machine learning models using the Gas Sensor Array Drift Dataset. By employing a chronological evaluation protocol rather than random train-test splits, the experiments preserved the temporal structure of the data to assess performance under realistic deployment conditions across a multi-month period. Five classification models representing distinct learning paradigms were evaluated under both static deployment (trained once) and adaptive deployment (periodically retrained).

The experimental results demonstrate that unmitigated temporal drift severely degrades predictive performance. Under static learning, all classifiers experienced substantial reductions in Accuracy, Balanced Accuracy, and Macro F1 by the final evaluation batch, with decision tree ensembles (Random Forest and XGBoost) suffering the most catastrophic performance collapse. Support Vector Machines and Multi-Layer Perceptrons exhibited greater baseline resilience but remained vulnerable to long-term performance decay.

A key insight of this study is that global centroid displacement in PCA space is not a reliable predictor of model failure. Batches with large centroid shifts often remained highly predictable, whereas batches with smaller global displacement frequently caused severe classification degradation. Latent space visual

analysis revealed that predictive performance depends primarily on whether class distributions maintain local separability within historical feature bounds, demonstrating that the structural geometry of drift is far more critical than its overall magnitude.

Furthermore, performance degradation was amplified by prior probability shift. Class proportions changed substantially over time; most notably Gas 6, which expanded from 2.2% in the initial training set to over 20% in later batches. These shifts caused severe class-specific failures when combined with feature drift, confirming that performance degradation arises from the joint effect of changes in feature distributions and changes in class proportions rather than from either phenomenon in isolation.

Periodic retraining proved highly effective in counteracting these failure modes. By continuously updating training support with newly observed batches, adaptive learning expanded feature space density coverage and realigned prior class weights. Consequently, all models achieved higher robustness scores and lower degradation rates. Notably, the performance gap between linear and complex non-linear architectures narrowed significantly, suggesting that access to representative and up-to-date data may be more important than model sophistication when operating in non-stationary environments.

From a practical perspective, these findings emphasize that long-term reliability in evolving systems, such as gas sensor arrays or HPC monitoring pipelines, requires active data management alongside model selection. Sustainable deployment strategies must incorporate continuous monitoring for both feature and prior shifts alongside structured retraining mechanisms.

Overall, the study highlights that robustness to temporal drift is fundamentally a data adaptation problem. While model architecture influences resilience to distribution changes, maintaining alignment between the training distribution and the evolving deployment environment is the key factor for sustaining long-term predictive performance.”

6 Links to external resources

The code and dataset used for this analysis are available in a dedicated GitHub repository: <https://github.com/davide-leone/gas-sensor-drift>.

7 LLM Usage Declaration

LLMs were used for code debugging, language polishing, and refactoring; all methodological choices, experiments, and interpretations were reviewed and validated by the author.